

Setup -- Lab 3 (Big Data Processing -- EPF)

Read and execute this document BEFORE the lab session.

If your stack is not running at the beginning of the lab, you risk to start the lab 30 min late (and we have a lot of things to learn together, I promise haha).

0. What you will build

By the end of this setup you will have:

- Docker Desktop running on your machine
- A folder `lab3_student/` extracted somewhere convenient
- Six containers running locally: Airflow scheduler, webserver, worker, triggerer, Postgres, Redis
- The Airflow UI loading at <http://localhost:8080>

Total disk usage: **~3 GB** (Docker images). Total RAM usage when running: **~3 GB**.

1. Prerequisites

Requirement	How to check
64-bit OS (Windows 10/11, macOS 12+, or Linux)	<code>systeminfo</code> (Windows), <code>uname -a</code> (Unix)
At least 8 GB RAM	Task Manager / Activity Monitor
At least 10 GB free disk	<code>df -h</code> or File Explorer
Admin rights on your machine	Required to install Docker

2. Install Docker Desktop (if you have Docker Desktop / Docker on your machine, you can skip sections 2 & 3)

Pick your OS:

Windows

1. Download Docker Desktop from <https://www.docker.com/products/docker-desktop/>.
2. Run the installer. **Tick the "Use WSL 2 instead of Hyper-V" option** when prompted.
3. Restart your machine.
4. Open Docker Desktop. Wait until the whale icon in the system tray stops animating.
5. Open **Settings** → **Resources** → **Advanced**:
 - CPUs: at least 2
 - Memory: at least **4 GB**
 - Disk image size: 32 GB is plenty
6. Apply & Restart.

macOS

1. Download Docker Desktop for Mac (Apple Silicon or Intel, pick the right one).
2. Drag to Applications and launch.
3. Same Resources settings as above.

Linux

1. Install Docker Engine + Docker Compose plugin: <https://docs.docker.com/engine/install/>.
2. Add your user to the `docker` group:

```
sudo usermod -aG docker $USER
newgrp docker
```

3. Log out and back in.

3. Verify Docker

Open a terminal (PowerShell on Windows, Terminal on Mac/Linux) and run:

```
docker --version
docker compose version
docker run --rm hello-world
```

The last command should print a paragraph starting with *"Hello from Docker!"*. If anything fails, fix it now, the lab cannot start without a working Docker.

4. Get the lab files

You will receive `lab3_student.zip` from the instructor (Moodle).

1. Download `lab3_student.zip`.
2. Extract it to a folder with a **short path and no spaces**:
 - **Good:** `C:\dev\lab3` (Windows), `~/lab3` (Mac/Linux)
 - **Bad:** `C:\Users\Jean-Pierre Dupont\OneDrive\My Documents\Big Data\Lab 3`
3. Open a terminal in that folder. From now on, all commands in the lab subject are run from that directory.
4. Create your local `.env` from the template:
 - Linux / macOS: `cp .env.example .env`
 - Windows (PowerShell): `copy .env.example .env`
5. **Linux and macOS - set `AIRFLOW_UID` to a number.** Docker Compose does **not** expand `$(id -u)` inside `.env`. If that text ends up in the file, you get: *Error response from daemon: unable to find user \$(id -u): no matching entries in passwd file*. After step 4, run one of:

```
sed -i "s/^AIRFLOW_UID=.* /AIRFLOW_UID=$(id -u) /" .env      # Linux (GNU
sed)
sed -i ' ' "s/^AIRFLOW_UID=.* /AIRFLOW_UID=$(id -u) /" .env  # macOS (BSD
sed)
```

Alternatively, open `.env` and set `AIRFLOW_UID=` to the integer printed by `id -u` (often `1000`).

6. **Windows (Docker Desktop):** keep `AIRFLOW_UID=50000` from `.env.example`; no change needed.

Long paths and spaces in paths **will create problems with Docker bind mounts** on Windows. If you cannot move out of `OneDrive`, at least make a junction: `mklink /J C:\dev\lab3 "C:\Users\you\OneDrive\...\lab3_student"`.

5. Pre-pull and build (could take some time if your internet is slow)

In the `lab3_student/` folder:

1. Pull only the **public** images (Postgres + Redis). The Airflow image for this lab is **built locally** from the `Dockerfile` (it adds DuckDB, pandas, pyarrow - all needed Python packages for the lab). If you run `docker compose pull` without building, you would get the stock Apache Airflow image and DAGs that import `duckdb` will fail at parse time.
2. Build the lab Airflow image:

```
docker compose pull postgres redis
docker compose build
```

The first `build` downloads the official Airflow base layer once, then installs the extra Python packages. Once finished, images are cached locally and the lab will start in seconds.

If you ever see `ModuleNotFoundError: No module named 'duckdb'`, run `docker compose build` (or `docker compose build --no-cache`) and then `docker compose up -d --force-recreate`.

6. First-time database initialization

Still in `lab3_student/`:

```
docker compose up airflow-init
```

This creates the Airflow metadata DB and the default admin user. Wait until you see a line like:

```
airflow-init-1 | User "airflow" created with role "Admin"
airflow-init-1 exited with code 0
```

Then bring the rest up:

```
docker compose up -d
```

Wait 60–90 seconds, then run:

```
docker compose ps
```

You should see **all containers in state Up or healthy**.

7. Verify the UI

Open <http://localhost:8080> in your browser.

Username	Password
airflow	airflow

You should land on the Airflow DAGs list. You will see a few example DAGs.

If the page does not load, run `docker compose logs airflow-webserver | tail -50` and check the output for errors.

8. Stop everything cleanly (for now)

You don't need Airflow running until the lab starts. Stop it without losing state:

```
docker compose stop
```

To start again later (this is the command you will run at the beginning of the lab):

```
docker compose start
```

To wipe everything and start from scratch:

```
docker compose down -v # -v also deletes the postgres volume
```

9. Common pitfalls

Symptom	Fix
Port 8080 already in use	Edit <code>.env</code> , change <code>AIRFLOW_UI_PORT=8081</code> , then <code>docker compose up -d</code>
<code>docker compose up</code> exits immediately with permission errors on Linux	Forgot to add yourself to the <code>docker</code> group (see §2 Linux)
Web UI loads but DAGs never appear	Scheduler is not running. <code>docker compose logs airflow-scheduler</code>
<code>airflow-init</code> keeps exiting with code 1	Probably an existing dirty Postgres volume. <code>docker compose down -v</code> and retry
Containers are unhealthy with "no space left on device"	Docker disk image is full. Settings → Resources → Disk in Docker Desktop, increase it
WSL2 takes 100 % CPU on Windows after a few hours	Known issue. Create <code>%UserProfile%\wslconfig</code> with <code>[wsl2]\nmemory=4GB\nprocessors=4</code> and reboot

10. What to bring on lab day

- Your laptop with this setup **already working** (i.e. you have completed §1 to §7).
- The `lab3_student/` folder extracted on a local disk.
- Your partners.

11. Help, I'm stuck

Email **theo.burdinat1@epfedu.fr** with:

- Your OS and Docker Desktop version
- The exact error message (copy-paste, not screenshot)
- The output of `docker compose ps` and `docker compose logs --tail=50`

Help requests sent too late may not be answered in time. Please do this setup in advance and make sure to have only one computer ready per group.

See you soon !